



भारतीय प्रौद्योगिकी संस्थान कानपुर
Indian Institute of Technology Kanpur
Department of Earth Sciences



Placement Brochure

Earth Sciences 2026-2027

Our Vision

“The BS-MS and M Tech. students of the Department of Earth Sciences, IIT Kanpur get through academic training that includes course works on fundamental and applied Earth Sciences as well as hands-on training on analytical techniques relevant to our society. Particularly, they learn about natural resources (identification and exploration techniques), Solid Earth Geology, Quaternary Geology, Sedimentology and Basin analysis, Natural Hazards, and Applied Geochemistry, Remote Sensing and Image Processing etc. Our students are well-trained with the theoretical aspects of hydrocarbon exploration from the basic to advanced stages, theoretical and practical Rock Mechanics, Rock Physics, and different Geophysical Exploration techniques with laboratory-based components. They visit several exploration industries, their mining sites, and beneficiary plants. In our teaching program, field geology is one of the essential components with emphasis on geological mapping and interpreting geological structures. IIT Kanpur campus environment encourages students to develop as better human beings, leaders, and team members.”

DR. SANTANU MISRA
PROFESSOR AND HEAD,
DEPARTMENT OF EARTH SCIENCES

The Earth Sciences Edge...

Department of Earth Sciences

IIT Kanpur



About us

The Department of Earth Sciences at IIT Kanpur focuses on the comprehensive study of the Earth, including its evolution, internal dynamics, surface and subsurface processes. It emphasizes understanding both natural and human-induced transformations of the terrestrial environment in the context of sustainable development—especially in light of the growing challenges posed by rapid population growth.

Lead Educators

Department of Earth Sciences | IIT Kanpur



Dr. Santanu Misra

Head of the Department (Professor)

Ph.D., Jadavpur Univ., 2007

[\[Homepage\]](#)



Dr. Javed N. Malik

Professor

Ph.D., M.S. Univ. Baroda, 1998

[\[Homepage\]](#)



Dr. Rajiv Sinha

Professor

Ph.D., Univ. Cambridge, 1992

[\[Homepage\]](#)



Dr. Debajyoti Paul

Professor

Ph.D., Cornell Univ., 2002

[\[Homepage\]](#)



Dr. Tajdarul H. Syed

Professor

Ph.D., UC Irvine, USA, 2007

[\[Homepage\]](#)



Dr. Deepak Dhingra

Associate Professor

Ph.D., Brown Univ., USA, 2014

[\[Homepage\]](#)



Dr. Dibakar Ghosal

Associate Professor

Ph.D., IGP, France, 2013

[\[Homepage\]](#)



Dr. Animesh Mandal

Associate Professor

Ph.D., IIT Kharagpur, 2013

[\[Homepage\]](#)



Dr. Indra S. Sen

Associate Professor

Ph.D., Florida Int. Univ., 2010

[\[Homepage\]](#)



Dr. Amar Agarwal

Associate Professor

Ph.D., IITR & KIT Germany, 2015

[\[Homepage\]](#)



Dr. Ishwar Kumar Cukkemane

Assistant Professor

Ph.D., IISc Bangalore, 2015

[\[Homepage\]](#)



Dr. Hiranya Sahoo

Assistant Professor

Ph.D., Univ. New Orleans, 2013

[\[Homepage\]](#)



Dr. Rabiul H. Biswas

Assistant Professor

Ph.D., PRL Ahmedabad, 2012

[\[Homepage\]](#)



Dr. Anupam Banerjee

Assistant Professor

Ph.D., IISc Bangalore, 2018

[\[Homepage\]](#)



Dr. Boddepalli Govindarao

Assistant Professor

Ph.D., IIT Kharagpur, 2019

[\[Homepage\]](#)



Dr. Deepa M. Veedu

Assistant Professor

Ph.D., NTU Singapore, 2019

[\[Homepage\]](#)



Dr. Thupstan Angchuk

Assistant Professor

Ph.D., JNU New Delhi, 2020

[\[Homepage\]](#)



Dr. Shyam Nandan

Assistant Professor

Ph.D., ETH Zurich, Switzerland, 2017

[\[Homepage\]](#)



Dr. Sumit Chakraborty

Visiting Faculty

Distinguished Professor

ALUMNI

Guiding Alumni

Our Alumni have been constant source of support and inspiration comprising highly knowledgeable and skilled individual of their field



Ashee Shukla
ESG,
SES Satellites, Luxembourg



Shobit Singh
Co-Founder & Director,
Terraqua UAV solutions pvt ltd



Aditya Parasramka
Product Manager,
INDmoney



Vani Chaturvedi
Assistant Manager Officer,
Bank of America



Abhishek Kumar
Sr. Data Scientist Analyst,
Accenture



Prateek Chougule
Project manager,
Mati Carbon



Mohammed Ali Khan
Research and Innovation
Engineer,
Adrosonic



Abhinav Uniyal
Project Manager,
GSI, India



Prabhati Sen
Geoscientist,
Schlumberger Limited



Nishant Jain
Data Engineer,
Mastercard



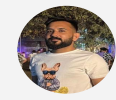
Tanmaya Naik
Analytics Manager,
Ola



Hammad Zubair
Geoscientist,
Baker Hughes



Vinay Joshi
Geoscientist,
IMC limited



Okesh Chaudhary
Senior Engineer,
Larsen and Toubro



Vaibhav Jain
Executive Geologist
ONGC, India



Saket Patidar
Petrophysicist,
Schlumberger Limited



Pranjal Pratap Dubey
Associate Product Manager
Pine Labs



Sudhanshu Pandey
Associate JP Morgan
Chase & Co.



Harshajit Borah
Geospatial Consultant,
ICF, India



Vedanth Venkatakrishnan
Consultant,
Bain & Company



Soumi Das
AI Engineer,
Fluxion Flowtech
Pvt. Ltd.

We

The Department of Earth Sciences blends rigorous field research with technical training. Students gain cutting-edge experience through geological fieldwork and hands-on advanced data analysis.

Thesis and project-based research is introduced early for both M.Tech and BS-MS students. This integrated approach prepares them for impactful careers in Academia, Research, and Industry.

Integrating Advanced Technical Skills into Research Excellence

Technical : Python, MATLAB, Machine Learning, SQL, Power BI, Tableau **Data**

Visualization

GIS & Remote Sensing: ArcGIS, QGIS, ENVI

Seismic & Geophysical Tools: Move Suite

Petrel, OpendTect

Geostatistics & Modeling: MATLAB,

Python

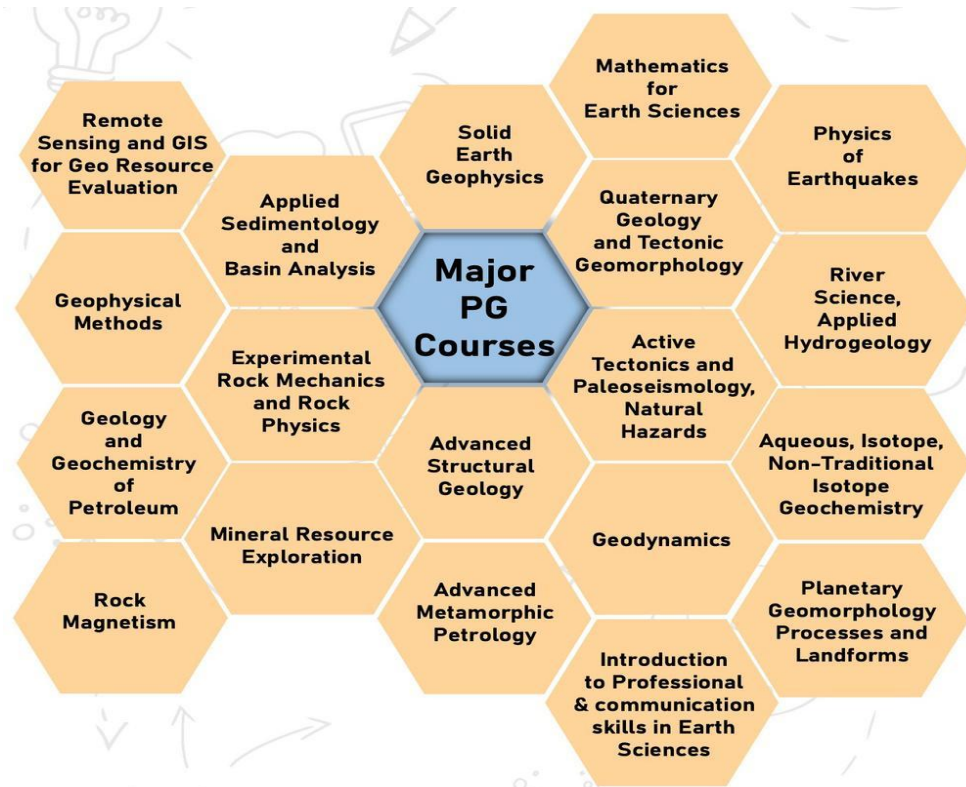


PG Course Curriculum

M.Tech Batch Strength: 10

Institute-level Courses Open to Earth Sciences Students

- Fluid Mechanics and Rate Processes
- Thermodynamics
- Probability and Statistics
- Fundamental of Computing
- Data Structure & Algorithm
- Introduction to Machine Learning
- Image Processing



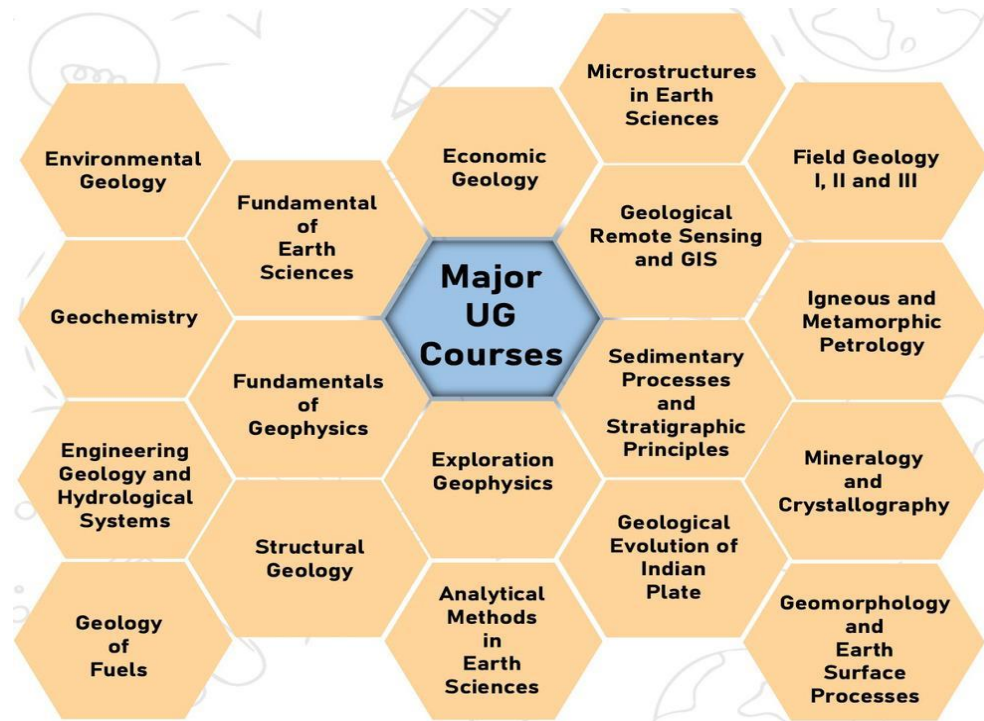
UG Course Curriculum

BS Batch Strength: 24

BS-MS Batch Strength: 5

Institute-level Courses Open to Earth Sciences Students

- Fluid Mechanics and Rate processes
- Thermodynamics
- Probability and Statistics
- Data Mining and Knowledge Discovery
- Fundamental of Computing
- Data Structure & Algorithm
- Introduction to Machine Learning
- Image Processing
- Macroeconomics
- Engineering Graphics



Advanced Instrumentation & Industry-Ready Skills

Workshop on WD-XRF and Stable Isotope Ratio Mass

Spectrometry (IRMS), '25

Workshop Coordinator: Prof. Debajyoti Paul

(Faculty Coordinator, XRF-IRMS Lab, ACMS, Department of Earth Sciences, IIT Kanpur)

Mastering Advanced Techniques:

Participants gained Practical expertise In High-precision analytical techniques such as **Wavelength Dispersive X-Ray Fluorescence (WD- XRF)** and **Stable Isotope Ratio Mass Spectrometry (IRMS)**. The sessions were designed to bridge the gap between academic learning and industrial application.

Access to Elite Equipment:

Hands-on training was provided using cutting-edge instruments including the **Rigaku Primus IV XRF** and the **Thermo Fisher DELTA Q IRMS**, housed at the **XRF-IRMS Lab, ACMS, IIT Kanpur**. These tools are crucial for advanced elemental and isotopic analysis across geoscience and environmental domains.

Solving Real-World Problems:

The workshop addressed impactful research areas such as **groundwater contamination** and **material composition analysis**, through expert sessions by **Prof. Abhas Singh** (Department of Civil Engineering, IIT Kanpur) and case studies rooted in real-world environmental challenges.

Forging Industry Partnerships:

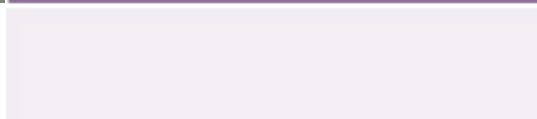
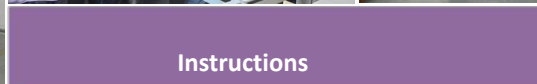
With collaborative support from **Rigaku** and **Thermo Fisher Scientific**, the workshop featured specialized talks and live demonstrations by **Mr.**

B.K. Srivastava (IR Tech, Rigaku) and **Dr. Piyush Deokar** (Thermo Fisher). These sessions ensured participants received industry-aligned technical exposure and insights into current instrumentation trends.

Hands-On



Instructions



Workshops

Guest Lectures – EAGE IIT Kanpur Student Chapter

As part of our mission to foster technical growth and industry awareness, EAGE IITK Student Chapter organized a series of insightful guest lectures featuring leading experts from academia and the energy sector. These sessions enriched students with knowledge of emerging geoscience techniques, reservoir evaluation, and real-world career perspectives.

FWI in Modern Exploration: Building Accurate Velocity Models for Imaging & Reservoirs

Dr. Saheb Ghosh – Superintending Geophysicist, GEOPIC, ONGC Dehradun

An applied lecture on Full-Waveform Inversion (FWI) technology in hydrocarbon exploration. The session covered FWI fundamentals, workflows, and a field case study demonstrating its superiority over conventional tomography in generating high-fidelity subsurface models for accurate drilling and reservoir delineation.

"Geoscience in the Energy Sector: Scope, Challenges, and Career Paths."

Dr. Satya Narayan – Senior Geophysicist, ONGC Dehradun

The scope and future of geoscience, including new technologies and trends. The role of geoscience in fossil fuel exploration and evolving techniques. An interactive discussion on internships, job roles, and career opportunities in the geoscience domain.

Magmatism, Hydrothermal Circulation, and Crustal Accretion

– Prof. Satish Singh (IPGP France)

Presented on Seismic Full Waveform Inversion at mid-ocean ridges. He explained how this geophysical technique reveals crustal formation and hydrothermal circulation across 60,000 km of plate boundaries, offering insights into deep-sea geological systems and the potential origins of life on Earth.

Well Logging and Formation Evaluation: Providing Insights for E&P Industry and Beyond

Dr. Ilus Mondal – Petrophysicist, bp Technical Solutions India (TSI)

Dr. Mondal explained the fundamentals of well logging and its vital role in subsurface exploration, detailing various logging tools. The session connected theoretical geoscience with high-impact exploration and production (E&P) applications, covering fluid saturation and log interpretation to identify hydrocarbon zones.



EAGE IITK SC
PRESENTS
GUEST LECTURE
BY

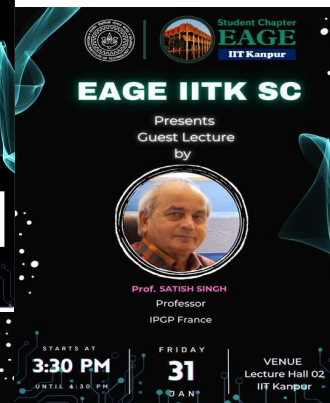
SAHEB GHOSH
SUPTDG. GEOPHYSICIST (S)
SPECIAL IMAGING SERVICES
PROCESSING DIVISION
GEOPIC, ONGC, DEHRADUN
STARTS AT 4:00 PM
UNTIL 6:00 PM
FRIDAY 09 DEC
Online MODE



EAGE IITK SC
Presents
Guest Lecture
by

DR. SATYA NARAYAN
Senior Geophysicist
ONGC, Dehradun
STARTS AT 11 AM
UNTIL 12 PM
SATURDAY 4 NOV
Online MODE

Hands-On



EAGE IITK SC
Presents
Guest Lecture
by

Prof. SATISH SINGH
Professor
IPGP France
STARTS AT 3:30 PM
UNTIL 4:30 PM
FRIDAY 31 JAN
VENUE Lecture Hall 02
IIT Kanpur



EAGE IITK SC
Presents
Guest Lecture
by

DR. ILIUS MONDAL
Petrophysicist
P&O Subsurface Disciplines
bp Technical Solutions India
(TSI)
STARTS AT 3:30 PM
UNTIL 6 PM
TUESDAY 10 DECEMBER
Online MODE



Gathering



Presenting



Research Scholars' Day '26

Research Scholars' Day '26 was a vibrant celebration of innovation, where scholars from the Department of Earth Sciences presented frontier research spanning seismology, geochemistry, tectonics, and environmental science. Held on February 8 at ESB-3 Auditorium, the event fostered dynamic exchanges between students, faculty, and distinguished guests from academia and industry, including **Dr. Waliur Rahaman** (NCPOR) and **Mr. Shameek Chattopadhyay** (SRK Consulting). **Research Scholars' Day '26** exemplified the department's commitment to research excellence, interdisciplinary collaboration, and industry-relevant problem solving.

Events

Field
Planning



Field Geology

Field geology is vital for geologists as it directly gives them experience (evidence) of both endogenic and exogenic processes. It allows them to observe present and past geological processes, collect first-hand data to interpret the dynamic processes of the Earth to uncover valuable insights about its history, and to evaluate prospectivity of natural resources.



Tool use



Evidence

Field Photos



Research Facilities

Sample preparation

Research facilities



Rock Cutting Machine, Abrismet 250, Bhuler USA



Desktop precision diamond wire saw with touch screen control NST-202A



Hot plate, Bain Geo Heat, METCO



High precision CNC PROXXON



Mounting Press



Low speed diamond saw, EQ-SYJ-150, MTI Corporation



Grinder, BOSCH



Vacuum oven Max temp. 300°C., LABTRO INC



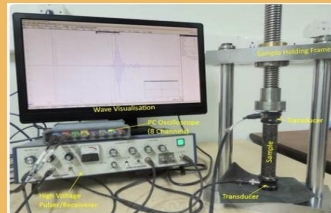
Vacuum Impregnation Unit



Thin Section preparation unit, PetroThin, Bhuler USA



Microwave sample preparator, Titan MPS, Parkin Elmer, Singapore



Longitudinal wave transducers, Blue Star Engg



Vibratory polisher, BUEHLER VibroMet 2



Double disc fully automatic grinder/polisher Bainpol PMV037, MATCO



Sanders, FERM

Physical/Mechanical Characterization

Research facilities



Compression Testing machine, Heico Engg



ESS1000 WITH 950-1000 Pound hammer (seismic energy source)



GEOPHONES, SG-10, Sercel S.A. France



RAU (CABLE FREE SEISMIC ACQUISITION UNIT), Sercel S.A. France



Portable magnetic susceptibility meter SM30, ZH Instruments



Portable magnetometer, MagDrone R3, SENSYS



VLF Instrument, GSMV 19 GEM Systems Inc. Canada



Quadcopter



Fixed wing drone. UX5 Arial Imaging rover, Trimble Europe



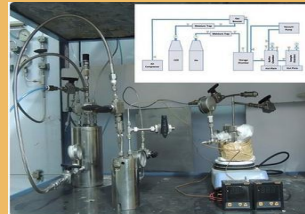
TX 8 scanner, Trimble Europe



Grain Size analyser, Sedigraph III-5120, Micromerits Instruments



Thermal, multispectral, hyperspectral sensors



High pressure temperature Adsorption Desorption setup



Triaxial load frame, WILLE GEOTECHNIK



Porosimeter-Permeameter, LP100A-GP101A, Porus Material Inc.

Chemical/Mineralogical/Petrographic



Polypropylene Fume Hood,
Polypropylene Nano Clean
Contamination control solutions



Vertical Laminar Airflow Fume
Hood, NU-164-424E, Medispec
Limited



Nutrient Analyser, XY-2 Sampler,
M/s Seal Analytical



Atomic Spectrometer, AAYLST 400,
Parkin Elmer, Singapore



X Ray Diffraction, Panlytical X-
PERT3, Panalytical



Cathode Luminescence MK5
electron gun, CITL MK5,
Cambridge Image tech



ICPMS, ICPQc, ThermoFisher
Scientific



Agilent® 8900 Triple Quad ICP-
MS



Ion Chromatography, 930
Compact IC Flex, Metrohm



Gamma Spectroscopy ORTEC
DSPEC jr 2.0



Liquid Water Isotope
Analyzer(LGR), LWIA-45-EP, Los
Getos Research, CA



Sunlight simulator, Honle
Evacube



Petrological microscope Leica



Scanning Electron Microscope
EM, JSM-6010PLUS/LA, JEOL
Asia Pte



Research grade polarising
microscopes from Leica and
Zeiss

Hire Us

1

Register

Register as an recruiter by Signing Up on the Recruitment Automation System Portal (RAS) by [Clicking here](#)

2

Fill Form

Complete the Job **Announcement Form (JAF) / Internship Proforma (IP)** on the afformtioned RAS portal.

3

Shortlist for Interviews

After reviewing resumes & portfolios - The company submits the list of **shortlisted students for the interview** process.

4

Select Students

Companies provide the list of **selected and wait-listed students** at the end of their interview slot.

5

Receive Acceptance

The Placement office notifies the companies regarding the **acceptance of selected students**.

6

Offer Letter

The company sends the **offer letter** to the selected candidates consistent with JAF or IP.

Placement

On & Off Campus



LARSEN & TOUBRO



Indian Mine Planners & Consultants
(A Geological, Mining & Environmental Consultants)





Contact

Student Placement Coordinators

Faculty In charge



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mridula25@iitk.ac.in



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+91-6295471168
pcharyya25@iitk.ac.in



Dr. Hiranya Sahoo

+91-5122592378
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Dr. Ishwar Kumar C.

+91-5122596952
ishwar@iitk.ac.in

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Find us: Department of Earth Sciences, IITK: [Website](#)

